

ECON 712: Applied Econometrics

Lecture 12: Statistical Properties of OLS, part II

Unbiasedness

- Under strict exogeneity, $\mathbb{E}[\mathbf{u}|\mathbf{X}] = \mathbf{0}$, OLS/MM is unbiased:

$$\hat{\beta}^{ols} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\hat{\beta}^{ols} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u}$$

$$\mathbb{E}[\hat{\beta}^{ols}|\mathbf{X}] = \beta \underbrace{=}_{\text{by LIE}} \mathbb{E}[\hat{\beta}^{ols}]$$

- Pre-determinedness, $\mathbb{E}[u_i|x_{1i}, \cdot, x_{ki}] = 0$, is not enough
- The linear CEF has to be correctly specified

Variance under Spherical Errors

- Another important property of OLS is the conditional **variance-covariance** matrix of estimates:

$$\text{Var}[\hat{\beta}^{ols}|\mathbf{X}] = \mathbb{E}[(\hat{\beta}^{ols} - \beta)(\hat{\beta}^{ols} - \beta)^T|\mathbf{X}]$$

- Assume that $\mathbb{E}[\mathbf{u}\mathbf{u}^T|\mathbf{X}] = \sigma^2 \mathbf{I}$, then

$$\text{Var}[\hat{\beta}^{ols}|\mathbf{X}] = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$$

- ▶ In practice, we estimate σ^2 by $\hat{\sigma}^2 = \hat{\mathbf{u}}^T\hat{\mathbf{u}}/(n - k)^{-1}$
- Spherical error terms is a very strict assumption, which is not usually satisfied, specially if the CEF is not linear
- By the law of total variance, the unconditional variance is

$$\text{Var}(\hat{\beta}^{ols}) = \sigma^2\mathbb{E}[(\mathbf{X}^T\mathbf{X})^{-1}]$$

Distribution of regressors under Gauss-Markov

- Classical or Gauss Markov Assumptions of Multiple Regression are:
 1. If the population equation is given by (linear CEF)

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

2. No perfect multicollinearity (i.e. $\mathbf{X}$ has full rank)
3. Zero Conditional Mean (i.e. $E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$)
4. Homoskedasticity and no serial correlation (i.e. $\text{Var}[\mathbf{u}|\mathbf{X}] = \sigma^2 \mathbf{I}_n$)
5. Population conditional error terms are jointly Normal:

$$\mathbf{u}|\mathbf{X} \sim N[\mathbf{0}, \sigma^2 \mathbf{I}_N]$$

- This yield the following result:

$$\hat{\boldsymbol{\beta}}|\mathbf{X} \sim N[\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}]$$

- Because of properties of the multivariate normal distribution:

$$\hat{\beta}_j|\mathbf{X} \sim N[\beta_j, \sigma^2 (\mathbf{X}^T \mathbf{X})_{jj}^{-1}]$$

- We can write the variance of the estimator in multiple regression as

$$\text{Var}[\hat{\beta}_j|\mathbf{X}] = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

Consistency

- Classical Assumptions are very restrictive and not usually satisfied in practice
- Therefore, modern econometrics relies on asymptotic properties of estimators, which are satisfied under weaker conditions
- For instance, if we assume that $\mathbb{E}[\mathbf{X}_i u_i] = \mathbf{0}$, then

$$\begin{aligned} \underset{n \rightarrow \infty}{plim} \hat{\beta}^{ols} &= \underset{n \rightarrow \infty}{plim} \left[\beta + \left(\frac{1}{n} \mathbf{X}^T \mathbf{X} \right)^{-1} \left(\frac{1}{n} \mathbf{X}^T \mathbf{u} \right) \right] \\ &= \beta + \left(\mathbf{S}_{\mathbf{X}^T \mathbf{X}} \right)^{-1} \underset{n \rightarrow \infty}{plim} \left(\frac{1}{n} \mathbf{X}^T \mathbf{u} \right) \\ &= \beta \text{ (the OLS estimator is consistent)} \end{aligned}$$

- Because we do not need mean independence, but zero covariance, the CEF does not need to be linear
- Pre-determinedness is enough

Asymptotic Distribution of OLS

- OLS decomposition:

$$\sqrt{n}(\hat{\beta}^{ols} - \beta) = \left(\frac{1}{n} \mathbf{X}^T \mathbf{X} \right)^{-1} \sqrt{n} \left(\frac{1}{n} \mathbf{X}^T \mathbf{u} \right)$$

- Under iid errors and $\mathbb{E}[X_i u_i] = 0$:

$$\frac{1}{n} \mathbf{X}^T \mathbf{X} \xrightarrow{p} S_{X^T X} \quad (\text{LLN})$$

$$\sqrt{n} \left(\frac{1}{n} \mathbf{X}^T \mathbf{u} \right) \xrightarrow{d} N(\mathbf{0}, \underbrace{\mathbb{E}[(X_i u_i)(X_i u_i)^T]}_{=\Omega}) \quad (\text{MCLT})$$

- By Slutsky:

$$\sqrt{n}(\hat{\beta}^{ols} - \beta) \xrightarrow{d} N(\mathbf{0}, S_{X^T X}^{-1} \Omega S_{X^T X}^{-1})$$

Asymptotic Variance of OLS

- We can write the matrix Ω as follows:

$$\Omega = \mathbb{E}[(\mathbf{X}_i u_i u_i)^T \mathbf{X}_i^T] = \mathbb{E}[u_i^2 \mathbf{X}_i \mathbf{X}_i^T]$$

- ▶ If $\mathbb{E}[u_i^2 | \mathbf{X}_i] = \sigma^2$, then by LIE

$$\mathbb{E}[u_i^2 \mathbf{X}_i \mathbf{X}_i^T] = \mathbb{E}[\mathbb{E}[u_i^2 | \mathbf{X}_i] \mathbf{X}_i \mathbf{X}_i^T]$$

$$\mathbb{E}[u_i^2 \mathbf{X}_i \mathbf{X}_i^T] = \sigma^2 S_{\mathbf{X}^T \mathbf{X}}$$

- ▶ This is the same variance as in the finite sample case, but with a large sample, we can set $S_{\mathbf{X}^T \mathbf{X}} = (\mathbf{X}^T \mathbf{X})$
- ▶ Then,

$$\sqrt{n}(\hat{\beta}^{ols} - \beta) \xrightarrow{d} N(\mathbf{0}, n \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1})$$

Heteroscedasticity-Robust Variance

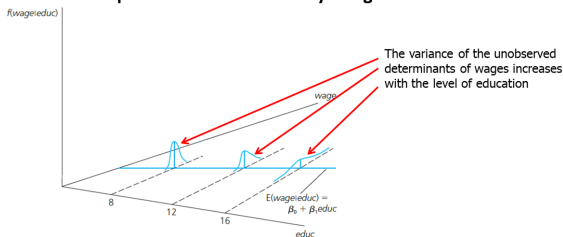
- Sometimes, error terms are non-spherical:

$$E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \Sigma_{\mathbf{u}} \neq \sigma^2 \mathbf{I}_N$$

- For instance, when there is heteroscedasticity

• An example for heteroskedasticity: Wage and education

$$\Sigma_{\mathbf{u}} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_N^2 \end{bmatrix}$$



Heteroscedasticity Robust Variance

- In this case, the Variance-Covariance Matrix estimated in a previous slide is incorrect:

$$\text{Var}[\hat{\beta}|\mathbf{X}] = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \Sigma_u \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}$$

- We could estimate Σ_u as follows:

$$\hat{\Sigma}_u = \begin{bmatrix} \hat{u}_1^2 & 0 & \cdots & 0 \\ 0 & \hat{u}_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{u}_n^2 \end{bmatrix}$$

- **HOWEVER**, this is not a consistent estimator as the number of parameters increases with n

Heteroscedasticity-Robust Variance

- We instead estimate the following term:

$$\mathbf{X}' \hat{\Sigma}_u \mathbf{X}$$

- ▶ This is a $k \times k$ matrix called a sandwich covariance matrix
- ▶ An element of the previous matrix is: $\sum_{i=1}^n \hat{u}_i^2 x_{is} x_{ij}$
- ▶ It can be shown that, under certain conditions, this is a consistent estimate (it biased in small samples as $E[\hat{u}_i^2 | \mathbf{X}] \neq \sigma_i^2$)
- In this case, the OLS estimator is not the most efficient estimator (FGLS is), but it is still used in practice

Cluster Robust Variance

- Suppose there is a natural grouping of data:

$$y_{ig} = \mathbf{x}_{ig} \beta + u_{ig}$$

or

$$\mathbf{y}_g = \mathbf{X}_g \beta + \mathbf{u}_g \quad \text{for } g = 1, \dots, G$$

where

$$\mathbf{X}_g = \begin{bmatrix} 1 & x_{12} & x_{13} & \cdots & x_{1k} \\ 1 & x_{22} & x_{23} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n_g 2} & x_{n_g 3} & \cdots & x_{n_g k} \end{bmatrix}$$

- The total number of observations is: $n = \sum_{g=1}^G n_g$

Cluster Robust Variance

- In addition, suppose that the variance structure of the error term is:

$$E[\mathbf{u}_g \mathbf{u}_g'] = \Sigma_g = \begin{bmatrix} \sigma^2 & \sigma_{1,2} & \cdots & \sigma_{1,n_g} \\ \sigma_{1,2} & \sigma^2 & \cdots & \sigma_{2,n_g} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1,n_g} & \sigma_{2,n_g} & \cdots & \sigma^2 \end{bmatrix}$$

and

$$E[\mathbf{u} \mathbf{u}'] = \Sigma_u = \begin{bmatrix} \Sigma_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \Sigma_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \Sigma_G \end{bmatrix}$$

Cluster Robust Variance

- In this case the sandwich estimator has the following form:

$$\mathbf{X}'\hat{\Sigma}_u\mathbf{X} = \sum_{g=1}^G \mathbf{X}'_g \hat{u}_g \hat{u}'_g \mathbf{X}_g$$

- Asymptotic Theory in this case applies for $G \rightarrow \infty$
 - ▶ A small number of groups result in large and very biased Se, independently of the size of n_g , specially when observations within groups are highly correlated
- It is always important to cluster if you suspect within group correlation, however, Clustering Se is extremely important in the following cases:
 - ▶ Time Series data with autocorrelated errors: $y_{it} = \mathbf{X}_{it}\beta + u_{it}$
 - ▶ Treatment level is higher than individual observations: $y_i = \mathbf{X}_g\beta + u_i$